

JEE ADVANCED - 7 | PAPER - 2

JEE 2022

Date 30th January, 2022	Maximum Marks 180	Duration 3 Hours
-----------------------------------	-----------------------------	----------------------------

General Instructions

- The question paper consists of 3 Subject (Subject I: **Physics**, Subject II: **Chemistry**, Subject III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
- Section 1** contains **2 types** of questions [**Type A** and **Type B**].
Type A contains **SIX (06) Multiple Correct Answers Type Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.
Type B contains **THREE (03) paragraphs**. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- Section 2** contains **6 Numerical Value Type Questions**. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
- For answering a question, an ANSWER SHEET (OMR SHEET) is provided separately. Please fill your **Test Code**, **Roll No.** and **Group** properly in the space given in the ANSWER SHEET.

Name of the Candidate (In CAPITALS) :

Roll Number :

OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

MARKING SCHEME

SECTION-1 | Type A

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is(are) chosen;
Partial Marks : +3 If all the four options are correct but ONLY three options are chosen;
Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks : 0 If unanswered;
Negative Marks : -2 In all other cases.
- For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then

choosing ONLY (A), (B) and (D) will get +4 marks;
 choosing ONLY (A) and (B) will get +2 marks;
 choosing ONLY (A) and (D) will get +2 marks;
 choosing ONLY (B) and (D) will get +2 marks;
 choosing ONLY (A) will get +1 mark;
 choosing ONLY (B) will get +1 mark;
 choosing ONLY (D) will get +1 mark;
 choosing no option(s) (i.e. the question is unanswered) will get 0 marks and
 choosing any other option(s) will get -2 marks.

SECTION-1 | Type B

- This section contains **THREE (03) paragraphs**. Based on each paragraph, there are **TWO (02)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : -1 In all other cases.

SECTION - 2

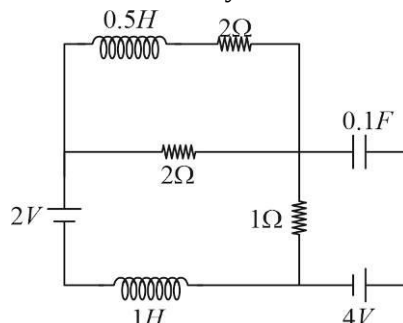
- This section contains **6 Integer Type Questions**. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If ONLY the correct Integer value is entered. There is **NO negative marking**.
Zero Marks: 0 In all other cases.

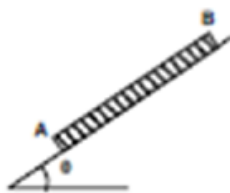
SECTION-1 | Type A

This section consists of 6 Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

1. The circuit shown in the figure has reached its steady state. Choose the correct options.

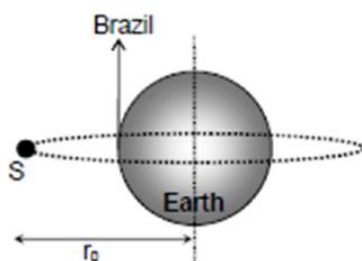


- (A) The charge on the capacitor is 0.1 C
 (B) The charge on the capacitor is 0.3 C
 (C) Power developed by the 2 Volts battery is 1 W
 (D) Power developed by the 2 Volts battery is 2 W
2. A uniform metal rod (AB) of mass m and length L is lying on a rough incline. The inclination of the incline is θ and coefficient of friction between the rod and the incline is μ . Chose the correct options.



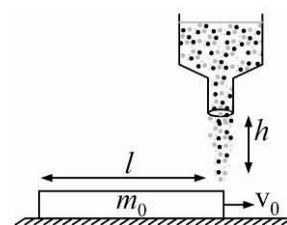
- (A) If temperature increases the rod expands. However, there is a point P on the rod which does not move. The distance of this point from the lower end of the rod is $\frac{L}{2} \left[1 + \frac{\tan \theta}{\mu} \right]$
 (B) If the temperature falls the rod contracts. Once again there is a point Q which does not move. The distance of Q from the lower end of the rod is $\frac{L}{2} \left[1 - \frac{\tan \theta}{\mu} \right]$
 (C) Repeated expansion and contraction cause the rod to slide down
 (D) Repeated expansion and contraction will not cause the rod to slide down
3. A Searle's experiment, a wire of diameter $d = 0.060\text{ cm}$ is used to find its young's modulus. The length of given wire is $L = 200\text{ cm}$ and when weight of mass $m = 30.0\text{ kg}$ is used in experiment, extension in the length of wire was found to be 0.200 cm . Then choose the correct option(s)
- (A) Maximum permissible error in young's modulus of wire is 4.67%
 (B) Maximum permissible error in young's modulus of wire is 3%
 (C) The least count of the instrument by which length of the wire is to be measured is 1 cm
 (D) The least count of the instrument by which diameter of the wire is to be measured is 0.001 cm

4. Consider a satellite 'S' which is just above Brazil and rotating on equatorial plane of earth. The sense of rotation of earth and satellite are similar (both are rotating from west to east). The time period of satellite is 6 hours. At $t = 0$, satellite is just above Brazil. Choose the correct option(s).



- (A) The time after which the satellite will be above Brazil again is 8 hours
 (B) The above satellite may be a geostationary satellite
 (C) The height of geostationary satellite above the sea level is 3580 km approximately
 (D) If the distance from the centre of earth to the satellite 'S' becomes $4r_0$, then the time after which the satellite will be above Brazil again is 48 hours

5. A plank of mass m_0 and length ℓ moving with velocity v_0 on a smooth horizontal surface passes under a stationary hopper. Sand spills from the hopper at a constant rate of μ kg/s, falls through height h onto the plank and sticks to it. Let v be the instantaneous velocity of plank, choose the correct option(s):



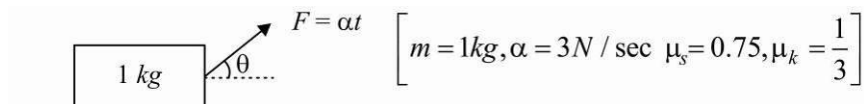
- (A) Horizontal component of force exerted by falling sand on the plank is proportional to v .
 (B) Vertical component of force exerted by falling sand on the plank is proportional to $\sqrt{h}$
 (C) Acceleration of the plank is proportional to v
 (D) Velocity of the plank after it completely passes from under the hopper is $v_0 e^{\frac{-\mu \ell}{m_0 v_0}}$
6. Choose the correct statement(s).
- (A) A particle of mass m at rest disintegrates into two particles of masses m_1 & m_2 . Let λ_1 & λ_2 are their de-Broglie wavelength respectively. It is given that $m_1 > m_2$ then $\lambda_1 > \lambda_2$.
 (B) The process of photoelectric emission is different to that of thermionic emission
 (C) Two radioactive samples A and B initially contain same number of nuclei. Sample A has half life of 1 year and B has half life of 2 years. At the end of 2 years, both samples have the same activity
 (D) When ultraviolet light is incident on a photocell, stopping potential is V_0 and the maximum kinetic energy of the photoelectrons is $K_{\max}$. When the ultraviolet light is replaced by x-ray. $K_{\max}$ increases but V_0 decreases

SECTION-1 | Type B

This section consists of THREE (03) paragraphs. Based on each paragraph, there are TWO (02) questions. Each question has FOUR options (A), (B), (C) and (D). ONLY ONE of these four options is the correct answer.

Paragraph for Question 7 - 8

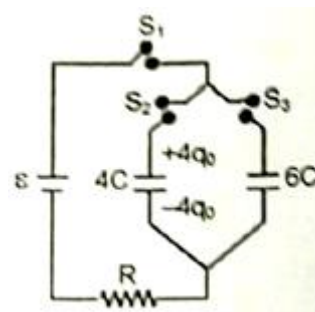
A variable force $F = \alpha t$ is acting on block of mass 'm' at an angle θ with horizontal as shown.



7. Minimum possible time after which block can start moving is :
 (A) 1sec (B) 2sec (C) 3sec (D) Data insufficient
8. Speed of block when it leaves the horizontal surface for $\theta = 37^\circ$ is approximately :
 (A) 20 m/s (B) 25 m/s (C) 28 m/s (D) 26 m/s

Paragraph for Question 9 - 10

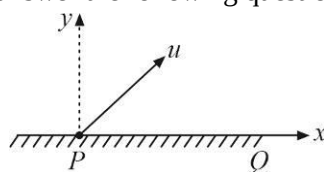
Two capacitors, one resistor and an ideal battery are connected through switches S_1, S_2 & S_3 as shown in the figure. Initially capacitor of capacitance $4C$ is charged, while other is uncharged. The upper plate of capacitor $4C$ is positive plate. Now, switches S_1, S_2 & S_3 are closed simultaneously then answer the following questions.



9. The current in resistor R in clockwise direction as a function of time is:
 (A) $\left(\frac{4C\epsilon - 2q_0}{5RC}\right)e^{-\frac{t}{10RC}}$ (B) $\left(\frac{5C\epsilon + 2q_0}{5RC}\right)e^{-\frac{t}{10RC}}$
 (C) $\left(\frac{5C\epsilon - 2q_0}{4RC}\right)e^{-\frac{t}{10RC}}$ (D) $\left(\frac{5C\epsilon - 2q_0}{5RC}\right)e^{-\frac{t}{10RC}}$
10. The charge on the capacitor of capacitance $6C$ as a function of time is :
 (A) $6C\epsilon - \left(6C\epsilon - \frac{12}{5}q_0\right)e^{-\frac{t}{10RC}}$ (B) $6C\epsilon - \left(6C\epsilon + \frac{12}{5}q_0\right)e^{-\frac{t}{10RC}}$
 (C) $6C\epsilon + \left(6C\epsilon - \frac{12}{5}q_0\right)e^{-\frac{t}{10RC}}$ (D) $6C\epsilon + \left(6C\epsilon + \frac{12}{5}q_0\right)e^{-\frac{t}{10RC}}$

Paragraph for Question 11 - 12

A small particle is projected from point P in a vertical plane with an initial speed u . It hits ground at Q . Assume that its collision with the ground which occurs at Q is perfectly inelastic. If area covered between the trajectory and x -axis by particle from P and Q is maximum, then answer the following questions.



11. Speed of particle at Q just after the collision is:
 (A) Zero (B) $\frac{u}{2}$ (C) $\frac{u}{\sqrt{2}}$ (D) $\frac{\sqrt{3}u}{2}$

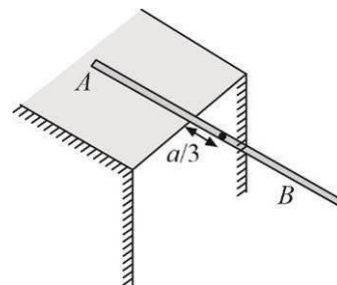
12. Displacement of particle from P to Q is:

(A) $\frac{\sqrt{3} u^2}{2 g}$ (B) $\frac{u^2}{2g}$ (C) $\frac{u^2}{\sqrt{2}g}$ (D) $\frac{u^2}{\sqrt{2}g}$

SECTION-2

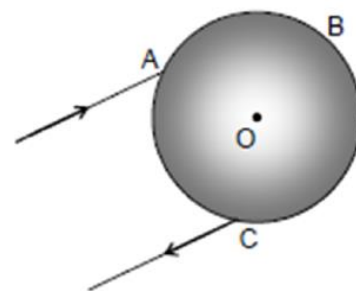
This section consists of 6 Numerical Value Type Questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08).

13. A rod AB of length $2a$ is kept horizontally on the edge of a table. Initially the centre of mass of the rod is at a distance $a/3$ from the edge of table. Now, if the rod is released from rest then rod starts sliding on the edge of table when it makes an angle θ from horizontal. If the coefficient of friction between rod and edge is $\lambda \tan \theta$, then find the value of λ .



14. A sphere of refractive index $\sqrt{2}$ is kept in a space whose centre is O . A ray is incident at point 'A' of the sphere and after refraction at A it is internally reflected at point 'B' then refracted at 'C' and finally comes out from the sphere. It is given that the total deviation produced in the incident ray is minimum, if the angle of incidence is given by

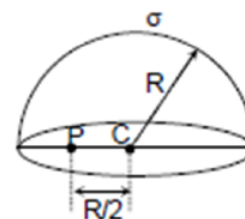
$i = \sin^{-1}\left(\sqrt{\frac{2}{\lambda}}\right)$, then find the value of λ .



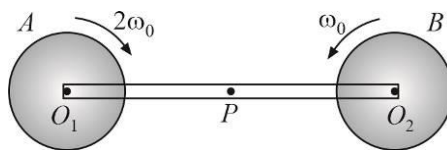
15. A sample of β active nuclei has a half-life of 100 minutes. Initially there are $10^{13}\beta$ active nuclei. Assume that all β particles emitted leave the sample. What is the charge acquired (in μC) by the sample in 200 minutes?
16. A particle is projected from ground with initial speed $v = \sqrt{30} \text{ m/s}$ and projection angle $\theta = 60^\circ$ from horizontal. If R is the radius of curvature of the trajectory of the particle when during flight particle's velocity make an angle $\frac{\theta}{2}$ with horizontal, then find the value of $R\sqrt{3}$ (Take $g = 10$).

17. Consider a uniformly charged hemispherical shell having charge per unit area σ . Find potential (in Volts) at point P due to the shell, R is the radius of shell. The distance of point P from the centre is $R/2$ (see diagram).

[Given $\frac{\sigma R}{\epsilon_0} = 4 \text{ SI units}$]



18. A rod of mass ' M ' and length $6R$ is pivoted at its centre (point P) to rotate in vertical plane as shown. Two discs of same radius ' R ' are joined at the ends O_1 and O_2 of the rod. Mass of the discs A and B are ' M ' and ' $2M$ ' respectively. Initial angular velocities of discs are shown in figure. The system is released from rest from horizontal position. The angular velocity of the rod when it is vertical is $\sqrt{\frac{g}{\lambda R}}$. Find the value of λ .



SECTION-1 | Type A

This section consists of 6 Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

- $\text{XeF}_6(\text{s})$ can be prepared by:

(A) $\text{Xe} : \text{F}_2$ in 1 : 20 mole ratio and high pressure (60 – 70 bar) and temperature 573 K

(B) $\text{Xe} : \text{F}_2$ in 1 : 5 mole ratio and pressure (7 bar) and temperature 873 K

(C) From product of (b), followed by an interaction with O_2F_2 at 143 K

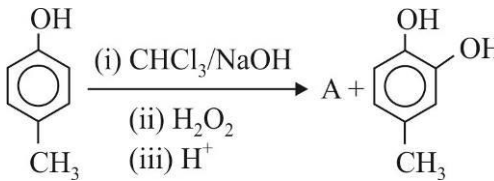
(D) Taking Xe in excess at atmospheric pressure (1 bar) and temperature (673 K)
- Which of the following methods can be used to produce phosphine gas?

(A) $\text{Ca}_3\text{P}_2 + \text{H}_2\text{O} \longrightarrow$

(B) $\text{PH}_4\text{I} + \text{KOH} \longrightarrow$

(C) Heating of H_3PO_3

(D) White P + NaOH $\xrightarrow[\text{atmosphere}]{\text{Inert}}$
- Which of the following species is/are planar with lone pair/s of electrons on the central atom?

(A) ClF_3 (B) XeF_5^- (C) PCl_5 (D) BrF_5
- 

(A) A will give +ve test with Tollen's reagent

(B) A is a weak organic monobasic acid

(C) A is an aldehyde

(D) A on reaction with conc. H_2SO_4 produces a neutral gas which is a very good reducing agent

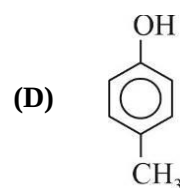
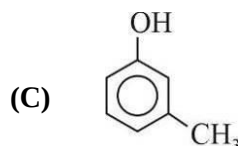
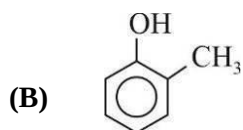
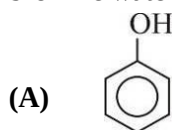
- Consider the following electrodes with their E° values:

$$E_{\text{Cu}^{2+}/\text{Cu}}^0 = +0.34\text{V}, E_{\text{Ni}^{2+}/\text{Ni}}^0 = -0.25\text{V}$$

$$E_{\text{Ag}^+/\text{Ag}}^0 = +0.8\text{V}, E_{\text{Zn}^{2+}/\text{Zn}}^0 = -0.76\text{V}$$

Which of the following reaction(s) is/are spontaneous?

- (A) $\text{Cu}(\text{s}) + \text{Ni}^{2+}(\text{aq}) \longrightarrow \text{Ni}(\text{s}) + \text{Cu}^{2+}(\text{aq})$
- (B) $\text{Cu}(\text{s}) + 2\text{Ag}^+(\text{aq}) \longrightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{Ag}(\text{s})$
- (C) $\text{Cu}(\text{s}) + 2\text{H}^+(\text{aq}) \longrightarrow \text{Cu}^{2+}(\text{aq}) + \text{H}_2(\text{g})$
- (D) $\text{Zn}(\text{s}) + 2\text{H}^+(\text{aq}) \longrightarrow \text{Zn}^{2+}(\text{aq}) + \text{H}_2(\text{g})$
- Which of the following compound(s) form(s) tribromo derivative(s) as the major product when treated with bromine water?

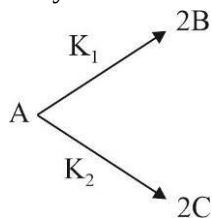


SECTION-1 | Type B

This section consists of THREE (03) paragraphs. Based on each paragraph, there are TWO (02) questions. Each question has FOUR options (A), (B), (C) and (D). ONLY ONE of these four options is the correct answer.

Paragraph for Question 7-8

For a hypothetical elementary reaction.



Where $\frac{K_1}{K_2} = \frac{1}{2}$, initially only 2 moles of A are present.

7. The total number of moles of A, B and C at the end of 50% reaction are:
 (A) 2 (B) 3 (C) 4 (D) 5
8. The ratio of number of moles of B to C at the end of 100% reaction is:
 (A) 2 : 1 (B) 1 : 2 (C) 1 : 3 (D) 2 : 3

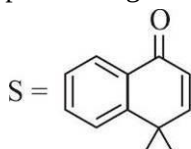
Paragraph for Question 9-10

The empirical formula of a cobalt (III) complex is $\text{Co}(\text{NH}_3)_x \text{Cl}_3$. The coordination no. of cobalt in the above complex is six. Assume that all complexes completely ionize in water.

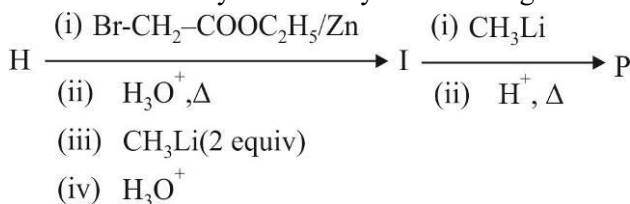
9. K_b of water is 2K kg mole^{-1} , when one mole of the above complex is added to 1kg water, the rise in boiling point is 6K. Then the value of x is:
 (A) 4 (B) 3 (C) 5 (D) 6
10. One mole of the above complex gave 143.5g white ppt with excess AgNO_3 solution. Then x is :
 [Mass number : Ag = 108, Cl = 35.5]
 (A) 4 (B) 5 (C) 6 (D) 3

Paragraph for Question 11-12

A 3° alcohol (P) upon acid catalysed dehydration gives a product Q. Ozonolysis of Q leads to only compound R which upon heating with KOH gives a compound S.

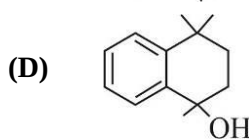
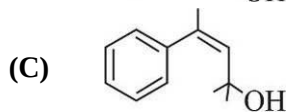
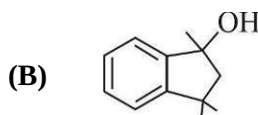
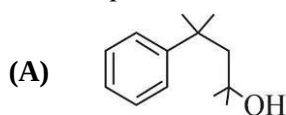


3° alcohol 'P' is synthesized by the following route:

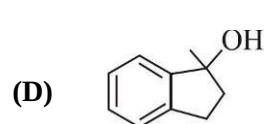
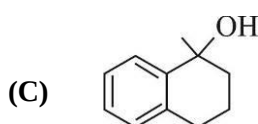
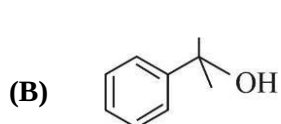
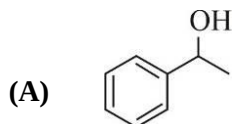


Molecular formula of H is $\text{C}_8\text{H}_8\text{O}$ and it gives positive iodoform test.

11. The compound 'P' is:



12. The compound 'H' on reaction with CH_3MgBr followed by hydrolysis gives:



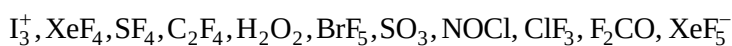
SECTION-2

This section consists of 6 Numerical Value Type Questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08).

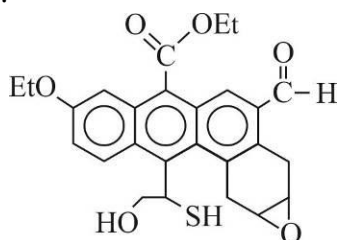
13. The oxidation state of metal in Fehling solution, on the reaction with benzaldehyde changes by units.

14. A decapeptide (Mol. wt. 796) on complete hydrolysis gives glycine (mol. wt. 75), alanine and phenylalanine. Glycine contributes 47% to the total weight of the hydrolysed products. The number of glycine units present in the decapeptide is:

15. Find the total number of species having planar shape.



16. How many moles of Grignard reagent are consumed in reaction of given compounds with excess of Grignard reagent?



17. Out of NiS , CoS , MnS , ZnS

A = Number of black precipitates

B = Number of sulphides which is/are precipitated by using H_2S in alkaline medium.

C = Number of sulphides which is/are soluble in CH_3COOH

D = Number of sulphides which is/are dissolved in aqua regia for confirmatory test.

What will be the value of $A + B + C + D$?

18. How many among the following species contain P – P bond(s):

(i) P (red)

(ii) $\text{H}_4\text{P}_2\text{O}_5$

(iii) $\text{H}_4\text{P}_2\text{O}_7$

(iv) $(\text{PO}_3^-)_3$

(v) $(\text{PO}_3^-)_2$

(vi) P_4O_{10}

(vii) P_4S_3

(viii) P_4O_6

(ix) P_4 (white)

SUBJECT III : MATHEMATICS

60 MARKS

SECTION-1 | Type A

This section consists of 6 Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

- If A and B are square matrices of order '3' such that $A \operatorname{adj}(2B) = 16I_3$ and $\det(B) = 2$ then which of the following is/are true (where $\operatorname{adj}(X)$ denotes adjoint of matrix (X) , X^{-1} denotes inverse of matrix (X) and $\det(X)$ denotes determinant value of matrix (X)).

(A) $(A^{-1}(\operatorname{adj}B))^{-1} = 4A^2$ (B) $(A^{-1}(\operatorname{adj}B))^{-1} = B^2$
 (C) $B(\operatorname{adj}A) = 8I_3$ (D) $B(\operatorname{adj}A) = 4I_3$
- Let $A(z_1)$, $B(z_2)$ and $C(z_3)$ lie on the circle $|z - i| = 1$ and satisfy the equation $3z_1 + i = 2z_2 + 2z_3$. If D is the centre of the circle $|z - i| = 1$, then which of the following is/are true.

(A) Area of quadrilateral $ABCD$ is $\frac{\sqrt{7}}{4}$ (B) Area of quadrilateral $ABCD$ is $\frac{\sqrt{11}}{4}$
 (C) $BC = \frac{\sqrt{7}}{2}$ (D) $BC = \frac{\sqrt{11}}{2}$
- Let $P_n = P_{n-1} + P_{n-2} \forall n \geq 2$ and $n \in N$ where $P_0 = 0$ and $P_1 = 1$, then which of the following is/are true:

(A) $\sum_{k=1}^{\infty} \frac{P_k}{3^k} = \frac{3}{5}$ (B) $\sum_{k=1}^{\infty} \frac{P_k}{3^k} = \frac{2}{5}$ (C) $P_9 = 34$ (D) $P_9 = 55$
- The probability that a married man watches a certain TV show is 0.4 and probability that a married women watches the show is 0.5. The probability that man watches the show, given that his wife does, is 0.7. Then which of the following is/are true:

(A) The probability that married couple watches the show is 0.45
 (B) The probability that a wife watches the show given that her husband does is $7/8$
 (C) The probability that married couple watches the show is 0.35
 (D) The probability that atleast one person of the married couple will watch the show is 0.55
- $P(x)$ is a fourth degree polynomial such that

(1) $P(-x) = P(x) \forall x \in R$ (2) $P(x) > 0 \forall x \in R$ (3) $P(0) = 1$
 (4) $P(x)$ has exactly two local minima at x_1 and x_2 such that $|x_1 - x_2| = 2$

The line $y = 1$ touches the curve at a certain point Q and the enclosed area between the line and the curve is $\frac{8\sqrt{2}}{15}$. Let $g(x) = Ax^2 + Bx + C$ ($A \neq 0$) such that $\lim_{x \rightarrow 0} \frac{P(x) - g(x) - g(-x)}{x^2}$ is finite and is equal to the slope of the tangent of $g(x)$ at $x = -1$. Also, $P(x)$ and $g(x)$ have common tangent at Q parallel to x -axis.

(A) The value of A is $-1/2$ (B) The value of $B + C$ is $-1/2$
 (C) The value of $A + C$ is 1 (D) The value of $A + B + C$ is -1
- All the three roots of $az^3 + bz^2 + cx + d = 0$ have negative real parts ($a, b, c, d \in R, a \neq 0$) then:

(A) $ab > 0$ (B) $bc > 0$ (C) $ad > 0$ (D) $a > 0$

SECTION-1 | Type B

This section consists of **THREE (03)** paragraphs. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for question 7 - 8

Let $S_r, r \in N$, denote the sum of an infinite GP whose first term is $2r - 1$ and common ratio $\frac{1}{2r}$. Let

$$V_n = \sum_{r=1}^n S_r \text{ and } Q_n = \sum_{r=1}^n \frac{1}{S_r S_{r+1}}.$$

7. $\lim_{n \rightarrow \infty} \frac{V_n}{n^2}$ equals:

- (A) 0 (B) -1 (C) 1 (D) 2

8. Consider the following statements:

I. $Q_1 > Q_2 > Q_3 > \dots$

II. $V_1 < V_2 < V_3 < \dots$

Then:

- (A) Only I is true (B) Only II is true
(C) Both I and II are true (D) Neither I nor II is true

Paragraph for question 9 - 10

Consider the function $f : R \rightarrow R$ defined by $f(x) = a^{-x} + a^{|x|}$, $a > 1$.

9. The interval in which $f(x)$ is monotonic decreasing, is:

- (A) $(0, a)$ (B) $(-\infty, 0)$ (C) $(0, \infty)$ (D) $(2, 2a)$

10. Area bounded by the given curve and the lines $x = -1$, $x = 1$ and $y = 0$, is:

- (A) $\frac{2a^2 + 3a + 1}{\log_e a}$ (B) $\frac{3a^2 - 2a - 1}{a \log_e a}$ (C) $\frac{3a - 1}{a \log_e a}$ (D) $\frac{(3a - 1)(2a - 1)}{\log_e a}$

Paragraph for question 11 - 12

Read the following write up carefully and answer the following questions:

Consider a function $f : R \rightarrow R$ such that $(f(x) + f(z))(f(y) + f(t)) = f(xy - zt) + f(xt + yz)$ where $x, y, z, t \in R$.

11. If $f : R \rightarrow R$ is the given function then in the interval $(0, \infty)$, f is:

- (A) Non decreasing (B) Non increasing
(C) Decreasing (D) None of these

12. The number of functions $f : R \rightarrow R$ which satisfy the given functional equation is:

- (A) 1 (B) 2 (C) 3 (D) None of these

SECTION-2

This section consists of 6 Numerical Value Type Questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08).

13. If $y = f(x)$ satisfies $f(x+1) + f(z-1) = f(x+z) \forall x, z \in R$ and $f(0) = 0$ and $f'(0) = 4$ then $f(2)$ is equal to ____.

14. Let $I_1 = \int_0^1 \frac{e^x}{1+x} dx$ and $I_2 = \int_0^1 \frac{x^2}{e^{x^3}(2-x^3)} dx$. The $\frac{I_1}{I_2 e}$ is equal to ____.

15. Let $f(x)$ be a real monic quadratic polynomial. If x_1, x_2, x_3, x_4, x_5 be the 5 points where $g(x) = |f(|x|)|$ is non-differentiable and $\sum_{i=1}^5 |x_i| = 8$ then $\frac{1}{5} \lim_{x \rightarrow \infty} \frac{x^2 - f(x)}{x}$ equals.

16. Let A and B are two sets such that

$$A = \{x : f(x) = [x^2], ([\cdot]) \text{ denotes GIF} \text{ is discontinuous in } [0, 2]\}$$

$$B = \{x : f(x) = \begin{cases} [\tan^2 x][\cot^2 x], & x \in (0, \pi) - \left\{\frac{\pi}{2}\right\} \\ 1 & x = \frac{\pi}{2} \end{cases} \quad ([\cdot]) \text{ denotes GIF} \text{ is non differentiable in } (0, \pi) \}$$

If m be the total number of onto function from A to B then total number of divisors of m is ____.

17. $\sum_{r=0}^9 {}^{20}C_{2r} {}^{20}C_{2r+2} = \frac{a}{a+1} ({}^{39}C_{21} - {}^{19}C_{10})$ and $xy = 4a$ then total number of ordered pair solution

(x, y) where $x, y \in I^+$ is ____.

18. Let set A contain all the (3×3) matrices whose entries are either 0, 1 or -1 . Two of these entries are 1, two

are -1 and five are 0. The number of matrices B that belongs to set A for which the equation, $B \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

has non-trivial solution is: